Anomaly detection for the monitoring of tree dieback from multispectral satellite images

The case of dark beetle infestations

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Outline

Bibliography and references

- This lecture is largely based on the presentation "Détection des problèmes sanitaires des forêts par imagerie satellite optique: principes et application au cas des scolytes de l'épicéa" by JB Féret, R Dutrieux, K Ose & F de Boissieu in October 4, 2022 at Atelier Theia (https://www.theia-land.fr/foret/2022-sante-foret/) and associated publication (https://www.onf.fr/onf/+/cec::les-rendez-vous-techniques-de-lonf-no69-70.html).

Bark beetle infestation



Figure: Bark Beetle. Source: Wikipedia

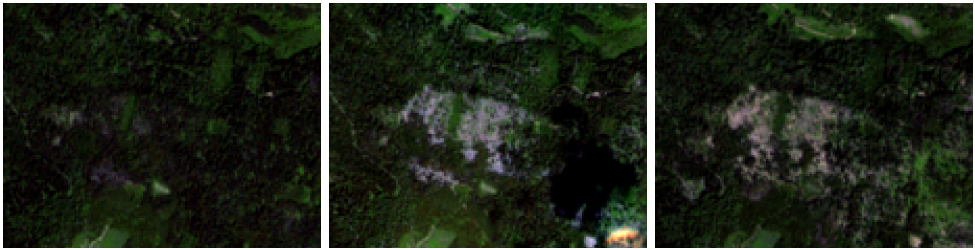


Figure: The architecture of the tunnel system varies by species. In the typograph moth, the egg-laying tunnels are generally vertical, while the larval tunnels are horizontal and winding. Source: Wikipedia[1]



Figure: Effects of bark beetle in the Bavarian Forest. Source: Wikipedia

[1] Meyers Konversations-Lexikon, 4. Aufl. 1888, Bd. 16, S. 352, Ausschnitt aus Tafel Waldverderber I (Käfer); Bd. 16, S. 352 aus Konversations-Lexikon im www[lien cassé], Domaine public, https://commons.wikimedia.org/w/index.php?curid=275883



(a) 2018 (b) 2019 (c) 2020

Figure: RGB composition of Sentinel-2 images (ESA, Copernicus) - Val d'Ajol, France

Let's start coding!